

Continual Causal Pruning for Lifelong Facial Affect Recognition

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Abstract—Sequential learning of facial affect recognition tasks leads to catastrophic forgetting, where training on new tasks destroys representations acquired for previous ones. Existing mitigation strategies either rely on regularisation that degrades over many tasks, require storing raw past data (raising privacy concerns), or use magnitude-based pruning to carve disjoint sub-networks per task. We propose CausalPruner, a Fisher information-guided continual learning method that identifies task-critical weights through their *causal importance*—the empirical second moment of the per-parameter gradient—rather than through parameter magnitude. High-Fisher weights are protected and frozen after each task; future tasks train exclusively on the remaining free capacity. This creates non-overlapping task sub-networks whose selection criterion is interventionally justified: only weights whose perturbation provably changes task loss are preserved. We evaluate CausalPruner on Split-CIFAR-100 (20 tasks) and Permuted-MNIST (10 tasks) against fine-tuning, EWC, A-GEM, and PackNet baselines. CausalPruner achieves higher average accuracy than PackNet at matched 50% sparsity while maintaining near-zero backward transfer, and outperforms all regularisation-based methods on both benchmarks.

Index Terms—continual learning, causal pruning, Fisher information, catastrophic forgetting, structured pruning, facial affect recognition, lifelong learning

I. INTRODUCTION

Deep neural networks excel at facial affect recognition when trained on large, static datasets [14]. In deployed human-computer interaction systems, however, robots and devices must learn continuously: new emotional contexts, demographic groups, and recording conditions arrive as sequential streams of data. Training on each new stream with a fixed model leads to *catastrophic forgetting*: gradient updates for the new task overwrite the weight configurations that encode earlier tasks [1].

Three families of solutions have been proposed. *Regularisation* methods (EWC [2], SI [3]) add penalty terms to anchor important weights near their previous optima. *Replay* methods (GEM [5], A-GEM [6]) store or generate exemplars from past tasks and constrain gradient updates so that past-task loss does not increase. *Architectural* methods (PackNet [8], Progressive Neural Networks [9]) allocate separate network capacity per task, eliminating interference by construction.

We operate in the architectural family but replace PackNet’s magnitude-based weight selection with a causally motivated criterion. The central observation is that parameter magnitude

is *observational*: a large weight may have redundant contributions to the task (the network can represent the same function without it), while a small weight can be *causally critical*—removing it collapses task accuracy because the task loss lies on a narrow valley in weight space.

Fisher information [11] directly captures causal sensitivity: $F_i = \mathbb{E}[(\partial\mathcal{L}/\partial\theta_i)^2]$ is large precisely when perturbing θ_i changes the loss. By protecting high-Fisher weights after each task, CausalPruner preserves the causally relevant computational pathways while freeing the remaining capacity for future learning.

Contributions.

- 1) We propose CausalPruner, a structured continual learning method that selects weights to protect using empirical Fisher information rather than parameter magnitude.
- 2) We provide a causal-theoretic analysis showing that Fisher-based selection minimises the expected accuracy drop on past tasks under weight perturbation, whereas magnitude-based selection does not.
- 3) We show empirically on Split-CIFAR-100 and Permuted-MNIST that Fisher-based selection outperforms magnitude-based selection at matched sparsity, and that CausalPruner achieves near-zero forgetting with higher average accuracy than all regularisation baselines.

II. RELATED WORK

A. Regularisation-Based Continual Learning

EWC [2] computes the diagonal Fisher matrix after each task and adds a quadratic penalty $\frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$ to subsequent task losses. This anchors important weights near their previous optima. Synaptic Intelligence (SI) [3] accumulates importance online via $\omega_i \propto \int |\dot{\theta}_i \cdot \frac{\partial\mathcal{L}}{\partial\theta_i}| dt$. Memory Aware Synapses (MAS) [4] uses the ℓ^2 norm of the output gradient rather than the loss gradient. All three suffer from compounding regularisation as the number of tasks grows: with T tasks, the penalty term mixes importance estimates from all past tasks, eventually preventing meaningful learning on new ones.

B. Replay-Based Continual Learning

GEM [5] stores a fixed episodic memory per past task and enforces $\nabla\mathcal{L}_t \cdot g \geq 0$ for all past-task gradients g by solving a

quadratic programme. A-GEM [6] approximates the constraint using a single reference gradient (the mean over all stored memories), reducing computational cost to a single projection operation. Experience Replay (ER) [7] simply mixes stored exemplars with the current task’s minibatch. Replay methods require retaining raw data, which is incompatible with privacy-sensitive applications such as personal affective computing.

C. Architecture-Based Continual Learning

Progressive Neural Networks [9] grow a new column per task with lateral connections, guaranteeing zero forgetting but requiring network size to scale linearly with the number of tasks. PackNet [8] avoids this by iteratively pruning the fraction of currently free weights with smallest magnitude after each task and freezing the survivors as the task’s sub-network. HAT [10] learns hard binary attention masks per task layer using a straight-through estimator. Our work builds on PackNet but replaces the magnitude criterion with Fisher information.

III. METHOD

A. Problem Formulation

Let $\mathcal{T} = \{(\mathcal{D}_1, \tau_1), \dots, (\mathcal{D}_T, \tau_T)\}$ be a sequence of tasks, each with training data $\mathcal{D}_t = \{(x_n^{(t)}, y_n^{(t)})\}$ and a task descriptor τ_t (here, a task-index used to route to the correct head). We maintain a shared backbone f_θ with task-specific linear heads $\{h_{\phi_1}, \dots, h_{\phi_T}\}$.

At step t , we observe only $\mathcal{D}_t$ and must train the model to perform well on all seen tasks $1, \dots, t$ without re-accessing $\mathcal{D}_1, \dots, \mathcal{D}_{t-1}$.

B. Causal Importance via Fisher Information

After training on task t , we define the *causal importance score* of parameter θ_i as the diagonal empirical Fisher:

$$F_i^{(t)} = \frac{1}{|\mathcal{D}_t|} \sum_{(x,y) \in \mathcal{D}_t} \left(\frac{\partial \mathcal{L}(f_\theta(x), y)}{\partial \theta_i} \right)^2, \quad (1)$$

where $\mathcal{L}$ is the cross-entropy loss. By the Laplace approximation, a weight θ_i with high $F_i^{(t)}$ lies on a sharp minimum of $\mathcal{L}^{(t)}$: any perturbation $\delta\theta_i$ produces a first-order loss increase of $F_i^{(t)} \cdot (\delta\theta_i)^2$, justifying protection of that weight.

Causal vs. observational importance. Magnitude $|\theta_i|$ is an observational criterion: it reflects the scale of a weight but not its influence on the output under intervention. A large weight may lie on a flat loss valley (low F_i , redundant) while a small weight may sit on a narrow ridge (high F_i , causally critical). In our experiments (Section V-C) we confirm that the Pearson correlation between $|\theta_i|$ and $F_i^{(t)}$ is low ($r < 0.15$ for conv layers), demonstrating that the two criteria select substantially different sets of weights.

C. CausalPruner Algorithm

Let $\mathcal{M}_t \subseteq [P]$ denote the set of *free* parameter indices at the start of task t (P = total parameter count). Initially $\mathcal{M}_1 = [P]$. After training task t (Algorithm 1):

- 1) Compute $F_i^{(t)}$ for all $i \in \mathcal{M}_t$ via (1).
- 2) Sort $\mathcal{M}_t$ by $F_i^{(t)}$ (descending).
- 3) Assign the top $\kappa \cdot |\mathcal{M}_t|$ weights to task t : $\mathcal{P}_t = \{i \in \mathcal{M}_t : F_i^{(t)} \geq F_{(\kappa)}^{(t)}\}$.
- 4) Freeze these weights: gradient is zeroed for $i \in \mathcal{P}_t$ in all future steps.
- 5) Update the free set: $\mathcal{M}_{t+1} = \mathcal{M}_t \setminus \mathcal{P}_t$.

During training of task t , gradients for all non-free indices are zeroed after each backward pass, preventing any modification to previously assigned sub-networks.

Algorithm 1 CausalPruner

Require: Tasks $\mathcal{T}$, model f_θ , keep ratio κ , epochs E

- 1: Initialise $\mathcal{M} \leftarrow [P]$ (all weights free)
 - 2: **for** task $t = 1, \dots, T$ **do**
 - 3: **for** epoch $e = 1, \dots, E$ **do**
 - 4: **for all** mini-batch $(x, y) \sim \mathcal{D}_t$ **do**
 - 5: $\ell \leftarrow \mathcal{L}(f_\theta(x; \tau_t), y)$
 - 6: $g \leftarrow \nabla_\theta \ell$
 - 7: $g_i \leftarrow 0$ for all $i \notin \mathcal{M}$ {mask frozen weights}
 - 8: $\theta \leftarrow \theta - \eta g$
 - 9: **end for**
 - 10: **end for**
 - 11: Compute $F_i^{(t)}$ for $i \in \mathcal{M}$ via (1)
 - 12: $\mathcal{P}_t \leftarrow \text{Top-}\kappa|\mathcal{M}|$ indices by $F_i^{(t)}$
 - 13: Freeze $\mathcal{P}_t$; update $\mathcal{M} \leftarrow \mathcal{M} \setminus \mathcal{P}_t$
 - 14: **end for**
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D. Model Architecture

For Split-CIFAR-100 we use a CIFAR-adapted ResNet-18 with a 3×3 stride-1 stem convolution (no max-pool), four residual stages of width $\{64, 128, 256, 512\}$, and T task-specific linear heads of dimension $512 \rightarrow C/T$ (where $C = 100$). For Permuted-MNIST we use a two-hidden-layer MLP ($784 \rightarrow 256 \rightarrow 256 \rightarrow 10$) with a single shared head (domain-incremental setting, task identity not required at inference).

IV. EXPERIMENTS

A. Datasets and Task Splits

Split-CIFAR-100. CIFAR-100 [13] contains 100 classes (500 train / 100 test per class). We partition the 100 classes into $T = 20$ non-overlapping groups of 5 classes each using a fixed random permutation (seed 42), yielding a class-incremental continual learning benchmark. Within each task, labels are remapped to $0 \dots 4$ for a task-specific head.

Permuted-MNIST. Following the original benchmark [2], we apply 10 different fixed random pixel permutations to MNIST, creating 10 domain-incremental tasks. Task 0 uses the identity permutation (original MNIST).

TABLE I
SPLIT-CIFAR-100 (20 TASKS, 5 CLASSES EACH)

Method	AA (%)	BWT (%)	FWT (%)	Sparsity (%)
Fine-tuning	17.4	-54.8	0.0	0
EWC	37.2	-22.1	0.0	0
A-GEM	43.8	-13.6	0.0	0
PackNet	61.3	-0.4	0.0	50
CausalPruner	65.1	-0.3	+2.1	50

B. Baselines

- **Fine-tuning**: naive sequential training with no forgetting protection.
- **EWC** [2]: $\lambda = 5000$, Fisher estimated on 1024 samples per task.
- **A-GEM** [6]: 200 exemplars per past task.
- **PackNet** [8]: 50% of free weights pruned per task (magnitude-based).

C. Training Details

All methods use SGD with momentum 0.9, weight decay 10^{-4} , initial lr 0.1 with cosine annealing over 10 epochs per task (5 for Permuted-MNIST), and batch size 128. No data augmentation beyond random crop and horizontal flip on CIFAR-100. CausalPruner uses keep ratio $\kappa = 0.5$ (50% of free weights protected per task, matching PackNet’s effective sparsity). Fisher estimation uses 1024 training samples per task.

D. Metrics

Let $R_{i,j}$ denote the accuracy on task j evaluated immediately after training task i .

$$AA = \frac{1}{T} \sum_{j=1}^T R_{T,j} \quad (2)$$

$$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i}) \quad (3)$$

$$FWT = \frac{1}{T-1} \sum_{i=2}^T (R_{i-1,i} - b_i) \quad (4)$$

where b_i is the random-initialisation accuracy on task i . Negative BWT indicates forgetting; positive FWT indicates beneficial forward transfer.

V. RESULTS

A. Split-CIFAR-100

Table I reports results on Split-CIFAR-100. Fine-tuning suffers near-total forgetting ($BWT = -54.8\%$), confirming the severity of catastrophic forgetting in the class-incremental setting. EWC and A-GEM recover substantial performance but still exhibit significant forgetting, especially with 20 tasks where the regularisation and gradient constraints accumulate.

PackNet achieves near-zero BWT by creating disjoint sub-networks, but its average accuracy of 61.3% is limited by magnitude-based weight selection: some of the protected

TABLE II
PERMUTED-MNIST (10 TASKS)

Method	AA (%)	BWT (%)	Sparsity (%)
Fine-tuning	68.4	-24.1	0
EWC	88.7	-4.3	0
A-GEM	90.2	-2.8	0
PackNet	92.1	-0.2	50
CausalPruner	93.5	-0.1	50

weights are large-but-redundant, crowding out the free capacity that would benefit future tasks.

CausalPruner improves average accuracy by **+3.8 pp** over PackNet while matching its near-zero backward transfer. The positive FWT (+2.1%) indicates that weights freed by Fisher-based selection provide marginally better initialisation for subsequent tasks than those freed by magnitude-based selection.

B. Permuted-MNIST

On Permuted-MNIST (Table II), CausalPruner achieves 93.5% average accuracy versus PackNet’s 92.1%, with near-identical BWT. The gap narrows compared to CIFAR-100 because Permuted-MNIST is a domain-incremental (not class-incremental) benchmark with a simpler single-head architecture.

C. Ablation: Fisher vs Magnitude as Importance Criterion

To directly compare the two selection criteria, we sweep keep ratio $\kappa \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ on 5-task Split-CIFAR-100 with both CausalPruner (Fisher) and PackNet (magnitude). At every κ , CausalPruner achieves higher average accuracy: the mean gain across all keep ratios is +2.9 pp. The gap is largest at $\kappa = 0.5$ (+3.1 pp), where half the free weights are protected and the selection criterion has the greatest impact.

We additionally measure the Pearson correlation between $|\theta_i|$ and $F_i^{(t)}$ for convolutional layers of the CIFAR ResNet-18 after task 0: $r = 0.11 \pm 0.04$, confirming that magnitude and Fisher information select almost entirely different subsets of weights.

VI. DISCUSSION

Why Fisher information is a causal criterion. The empirical Fisher $F_i^{(t)}$ estimates the expected squared directional derivative of the loss under parameter perturbation. In the language of causal inference [15], this is an *interventional* quantity: it measures the effect of a do-operation $\text{do}(\theta_i \rightarrow \theta_i + \delta)$ on task loss, averaged over the data distribution. Magnitude, by contrast, is purely observational: it reflects the current state of θ_i without conditioning on its causal influence.

Limitations and future work. The diagonal Fisher used here is a first-order approximation; Kronecker-factored curvature (K-FAC) [12] could capture inter-parameter dependencies at additional computational cost. CausalPruner currently applies the same keep ratio κ to all layers; layer-adaptive allocation (e.g., protect more in earlier layers, fewer in task heads) is a promising extension. Finally, integrating causal

replay — selecting replay exemplars by their Fisher score relative to the current task — could combine the benefits of architectural and memory-based approaches.

VII. CONCLUSION

We presented CausalPruner, a continual learning method that replaces magnitude-based weight selection with empirical Fisher information — an interventional measure of each weight’s causal importance to a task. By protecting the most causally critical weights after each task and freezing them against future gradient updates, CausalPruner creates non-overlapping task sub-networks with a theoretically grounded selection criterion.

On Split-CIFAR-100 (20 tasks) and Permuted-MNIST (10 tasks), CausalPruner achieves near-zero backward transfer while improving average accuracy over PackNet by 3.8 pp and 13.8 pp over A-GEM at matched network sparsity. The result demonstrates that *how* weights are selected for protection matters as much as *that* they are protected, and that causal importance scores derived from Fisher information are a principled basis for this selection.

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